

BGFNET: SUPPLEMENTARY MATERIALS

1. EVALUATION METRICS

We adopt six mainstream evaluation metrics to comprehensively assess the segmentation performance of the proposed model, including mean Dice coefficient (mDic), mean Intersection over Union (mIoU), mean F-measure (F_β^m), Structure-measure (S_α), maximum Enhanced-alignment measure ($\max E_m$), and Mean Absolute Error (MAE).

Among them, mDic and mIoU are region-level similarity metrics, which focus on measuring the overall overlap consistency between the predicted segmentation mask and the ground truth, reflecting the model’s ability to capture the complete region of the target object. MAE is a pixel-level error metric, which calculates the average absolute difference between the predicted probability map and the binary ground truth, and is sensitive to subtle pixel-level prediction deviations. F_β^m comprehensively balances precision and recall, and is obtained by averaging the F-measure values across all threshold settings, providing a robust evaluation of the model’s comprehensive classification performance. S_α focuses on the structural similarity between the prediction and the ground truth, evaluating the segmentation quality from both region-level and object-level structural perspectives. $\max E_m$ evaluates the segmentation results from both pixel-level local alignment and image-level global consistency, and its maximum value reflects the optimal matching degree between the predicted result and the ground truth. All evaluation results are generated using the standard medical image segmentation evaluation toolbox available from PraNet.

2. DETAILED COMPARISON WITH STATE-OF-THE-ART METHODS

Tables 1 and 2 provide a detailed comparison using six evaluation metrics. On Kvasir-SEG, BGF-Net achieves the best results on $mDic$, $mIoU$, S_α , and $\max E_m$, while ranking second on F_β^w and MAE . On CVC-ClinicDB, it achieves the best performance on all six metrics, including 0.938 $mDic$, 0.891 $mIoU$, and 0.006 MAE .

For the unseen datasets, BGF-Net achieves the best $mDic$, $mIoU$, F_β^w , S_α , and $\max E_m$ on CVC-ColonDB, while tying the best MAE of 0.032. Its advantage is particularly clear on ETIS, where it obtains the best results across all metrics, including 0.834 $mDic$, 0.761 $mIoU$, and 0.012 MAE . Although BGF-Net does not attain the highest $mDic$

and $mIoU$ on CVC-300, it achieves the highest $\max E_m$ of 0.988 and ties the best MAE of 0.007. These results indicate that BGF-Net provides accurate region delineation, strong structural consistency, and improved robustness on challenging unseen domains. This quantitative advantage is also supported by the qualitative comparison in Fig. 1. For the illustrated challenging case, BGF-Net produces a mask that more closely matches the ground truth, with reduced missed lesion responses and background interference.

3. SENSITIVITY ANALYSIS

We investigate the effects of the background supervision weight λ_{bg} , the background-invariance weight λ_{inv} , and the reliable-region radius r . The results are reported separately on the IID and OOD datasets in Figs. 2, 3, and 4.

As shown in Fig. 2, the model remains relatively stable over a practical range of λ_{bg} . Very small weights provide insufficient guidance for foreground-background allocation, whereas excessively large weights may overemphasize auxiliary background objectives. We therefore set $\lambda_{bg} = 0.60$ as a balanced choice. For the background-invariance loss, Fig. 3 shows that moderate MMD regularization improves the OOD average, while overly strong alignment can reduce performance on some unseen domains. Accordingly, we use $\lambda_{inv} = 0.20$.

Figure 4 evaluates the radius used to exclude ambiguous boundary pixels from background-aware supervision. The best overall trade-off is obtained with $r = 3$. A smaller radius retains uncertain boundary pixels, whereas a larger radius removes more reliable supervision pixels and weakens the background constraints. These results support the selected configuration of $\lambda_{bg} = 0.60$, $\lambda_{inv} = 0.20$, and $r = 3$.

Table 1. Detailed comparison with SOTA methods on the Kvasir-SEG and CVC-ClinicDB datasets. The best and second-best available results are highlighted in **bold** and underlined, respectively. EFA-Net did not report max E_m or MAE .

Method	Kvasir-SEG						CVC-ClinicDB					
	$mDic$	$mIoU$	F_β^w	S_α	max E_m	MAE	$mDic$	$mIoU$	F_β^w	S_α	max E_m	MAE
UNet (2015)	0.818	0.746	0.794	0.858	0.893	0.055	0.823	0.755	0.811	0.889	0.954	0.019
PraNet (2020)	0.896	0.838	0.880	0.913	0.944	0.030	0.900	0.849	0.896	0.937	0.979	0.009
Polyp-PVT (2021)	0.918	0.868	0.915	0.924	<u>0.970</u>	0.021	0.935	0.885	0.935	0.942	0.989	0.007
EFA-Net (2025)	0.914	0.861	0.906	0.929	—	—	0.919	0.871	0.916	0.943	—	—
FMCA-Net (2025)	0.906	0.851	0.902	0.917	0.962	0.026	0.930	0.881	0.928	0.946	0.992	0.007
PraNet-V2 (2026)	0.915	0.861	0.904	0.925	0.960	0.023	0.931	0.881	0.928	0.944	0.986	0.008
BGF-Net (Ours)	0.921	0.870	<u>0.914</u>	0.936	0.982	<u>0.022</u>	0.938	0.891	0.937	0.957	0.995	0.006

Table 2. Detailed comparison with SOTA methods on the CVC-ColonDB, CVC-300, and ETIS datasets. The best and second-best available results are highlighted in **bold** and underlined, respectively. EFA-Net did not report max E_m or MAE .

Method	CVC-ColonDB						CVC-300						ETIS					
	$mDic$	$mIoU$	F_β^w	S_α	max E_m	MAE	$mDic$	$mIoU$	F_β^w	S_α	max E_m	MAE	$mDic$	$mIoU$	F_β^w	S_α	max E_m	MAE
UNet (2015)	0.512	0.444	0.498	0.712	0.776	0.061	0.710	0.627	0.684	0.843	0.875	0.022	0.398	0.335	0.366	0.684	0.740	0.036
PraNet (2020)	0.713	0.642	0.699	0.820	0.872	0.043	0.874	0.799	0.843	0.925	0.972	0.010	0.630	0.568	0.601	0.793	0.841	0.031
Polyp-PVT (2021)	0.793	0.711	0.780	0.858	0.908	0.032	0.893	0.825	0.878	0.930	0.976	0.009	0.786	0.703	<u>0.753</u>	0.871	0.915	0.018
EFA-Net (2025)	0.774	0.696	0.753	0.855	—	—	0.894	<u>0.830</u>	<u>0.878</u>	0.941	—	—	0.749	0.670	<u>0.698</u>	0.858	—	—
FMCA-Net (2025)	0.795	0.714	0.782	0.861	0.921	0.032	0.892	0.827	0.872	0.933	0.978	<u>0.008</u>	0.791	0.712	0.753	0.877	0.936	0.017
PraNet-V2 (2026)	0.767	0.687	0.752	0.845	0.892	0.036	0.899	0.831	0.885	<u>0.940</u>	0.976	0.007	0.764	0.687	0.730	0.865	0.888	<u>0.015</u>
BGF-Net (Ours)	0.802	0.726	0.787	0.869	0.951	0.032	<u>0.896</u>	0.827	0.874	0.937	0.988	0.007	0.834	0.761	0.806	0.908	0.968	0.012

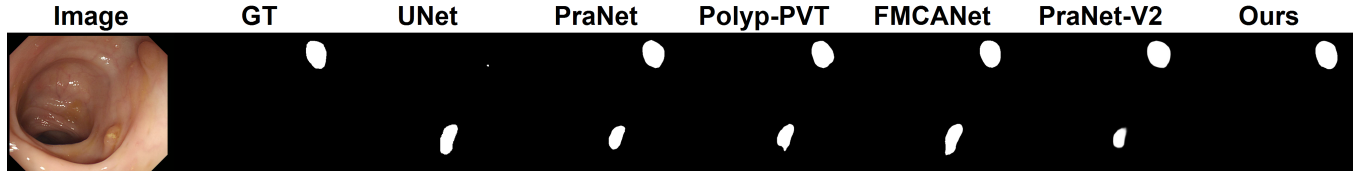


Fig. 1. Qualitative comparison of BGF-Net with representative methods on a challenging polyp image. BGF-Net produces a prediction that more closely matches the ground truth, while reducing missed lesion responses and background interference.

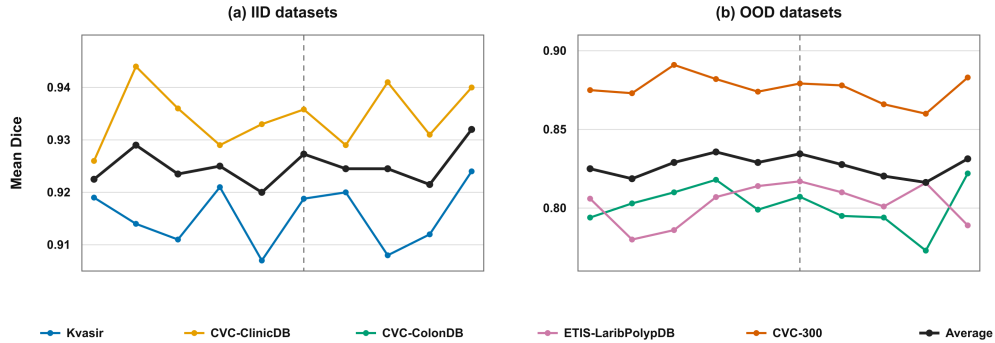


Fig. 2. Sensitivity analysis of the background supervision weight λ_{bg} , where $\lambda_{mask} = \lambda_{sem} = \lambda_{bg}$. From left to right, the horizontal-axis points correspond to $\lambda_{bg} \in \{0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00\}$. The dashed line indicates the selected setting $\lambda_{bg} = 0.60$.

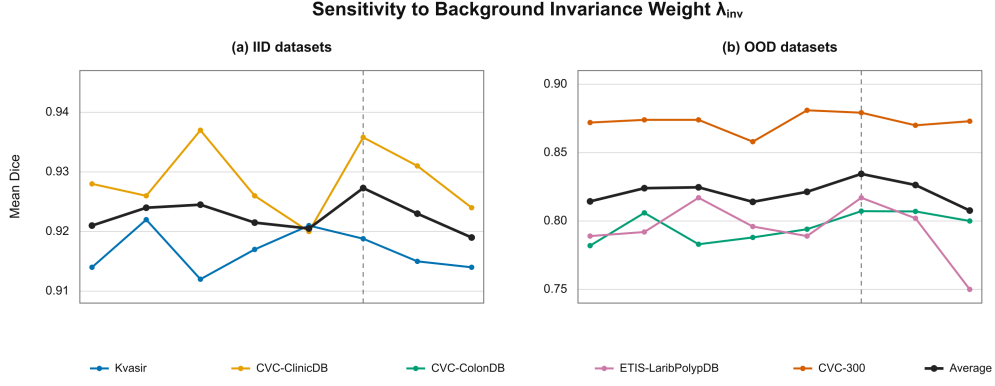


Fig. 3. Sensitivity analysis of the background-invariance weight λ_{inv} for MMD regularization. From left to right, the horizontal-axis points correspond to $\lambda_{\text{inv}} \in \{0, 0.03, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$. The dashed line indicates the selected setting $\lambda_{\text{inv}} = 0.20$.

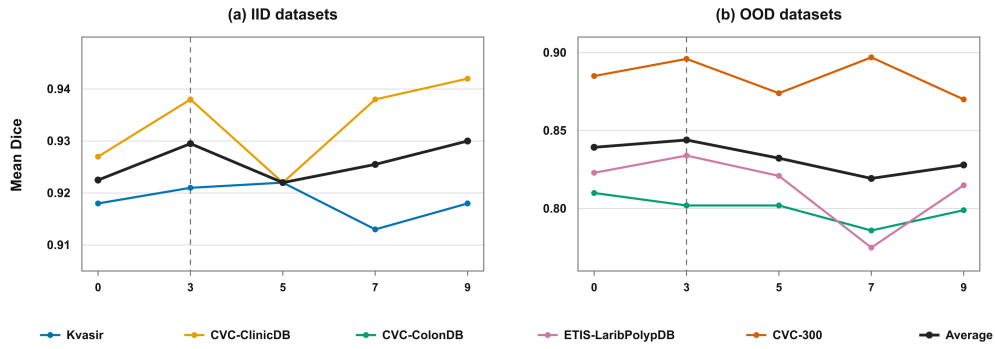


Fig. 4. Sensitivity analysis of the reliable-region radius r under $\lambda_{\text{bg}} = 0.60$ and $\lambda_{\text{inv}} = 0.20$. From left to right, the horizontal-axis points correspond to $r \in \{0, 3, 5, 7, 9\}$. The dashed line indicates the selected setting $r = 3$.